Department of Mathematics
Dhaka College, Dhaka
3rd year honors Examination-2023
Math Lab-II, FORTRAN (Assignment Questions)

1. A function $f(x)$ is defined by $f(x)=\begin{cases} -x & if \quad x\le 0 \\ x & if \quad 0<x<1 \\ 1-x & if \quad x\ge 1 \end{cases}$. Write a FORTRAN program segment to evaluate above function.

2. A function is defined as follows: $f(x)=\begin{cases} 1-x^2, \ x<0 \\ 1+2x, \ 0\le x<1 \\ 3+\dfrac{1}{x}, \ x>1 \end{cases}$ If $x\in[-10,10]$ then find the value of $f(x)$ by using FORTRAN for each $x=0.5$

3. Consider $f(x,y)=\begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2}, & When \quad (x,y)\ne(0,0) \\ 0, & When \quad (x,y)=(0,0) \end{cases}$ Write a FORTRAN program which find $f_{xy}(0,)$ and $f_{xy}(0,0)$ and print $f_{xy}(0,0)\ne f_{xy}(0,0)$.

4. Compute $\int_0^6 \dfrac{dx}{1+x^2}$ by simpson's $\dfrac{1}{3}$ rule and simpson's $\dfrac{3}{8}$ rule and compare your result to exact value.

5. Find the dominant eigenvalue and the corresponding eigenvector of the following matrix: $\begin{pmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \ with \ X^{(0)}=\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

6. Compute $\int_0^{\frac{\pi}{2}} e^{\sin x}dx$ corrected up to 5 decimal places by Simpson's $\dfrac{3}{8}$ rule and Trapezoidal rule. After finding the true value of the integral, compare the errors in these cases.

7. Write a FORTRAN program to find the sum of z, where $z=x^2-2xy+3y^2-8x+3y-8$ for the values of x and y range -3 to 3 with increment 0.2.

8. Solve the following system correct to five decimal places

$$2x+4y+6z=22$$
$$3x+8y+5z=27$$
$$-x+y+2z=2$$

by using LU factorization method. Also show L and U.

9. Solve the following system correct to five decimal places

$$2x+4y+6z=22$$
$$3x+8y+5z=27$$
$$-x+y+2z=2$$

by using SOR iterative method with three different values of ω .

10. For Romberg integration to approximate the value of the integral $\int_0^{\pi/4} x^2\sin x\,dx$ correct up to five decimal places. Compare your result with the exact value 0.0887553. Show your result in a tabular form.

11. Write a FORTRAN program, which prints the first 20 Fibonacci numbers and determine whether each number is each number is prime or not.

12. Write a FORTRAN program to find the product AB = C where A and B are defined as $A=\begin{pmatrix}1&3&5\\2&5&7\\3&8&1\end{pmatrix}$ $and\ B=\begin{pmatrix}3&2&15\\2&3&7\\13&8&10\end{pmatrix}$.

13. Use LU decomposition method to solve the system of linear equations: AX = B where $A=\begin{pmatrix}4&1&-1\\2&7&1\\1&-3&12\end{pmatrix}$, $X=\begin{pmatrix}a\\b\\c\end{pmatrix}$ $and\ B=\begin{pmatrix}3\\19\\31\end{pmatrix}$. Show the results of L and U and solutions.

14. the following series:

(i) $\left(1+\frac{1}{2}\right)+\left(1+\frac{1}{2}+\frac{1}{3}\right)+...+\left(1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{100}\right)$ (ii) $1+\frac{1}{2!}+\frac{1}{3!}+...+\frac{1}{25!}$; where $n\in N$

15. Write a FORTRAN program to find it output, to obtain the root of the equation $\cos x=3x-1$ correct to four decimal places, using fixed point iteration method.

16. Write a FORTRAN program for bisection method to find a root of the equation $e^{-x}-x=0$

17. Write a FORTRAN program to find the largest Eigen value of the matrix $\begin{pmatrix}5&0&1\\0&-2&0\\1&0&5\end{pmatrix}$

18. Use Newton's forward difference interpolation formula to evaluate F(2003) for the following set of tabulated values:

X	2000	2005	2010	2015	2020
Y=F(X)	100	125	139	200	250

19. Write a FORTRAN program which reads a 5×5 matrix and find its inverse matrix.

20. Write a FORTRAN program to find the product of AB, Where A and B are matrices of order $l\times m$ and $m\times n$ respectively.

21. Using Lagrange's interpolation formula of 2-points, 3-points and 4-points, write a program to estimate $f(6.0)$ employing the following data sets

x	1.0	4.0	9.0	16.0
$f(x)$	101.0	102.0	103.0	104.0

22. A class of 20 students takes the four test on computer course. Write a FORTRAN program which finds
    (i) The average score
    (ii) The no. of first class scores (i.e. score $\geq$ 60%)
    (iii) The no. of second class scores (i.e. score 45% $\leq$ score $\leq$ 60%)
    (iv) The no. of third class scores (i.e. score 40% $\leq$ score $\leq$ 45%)
    (v) The no. of those which failed.

23. Write a FORTRAN program to solve $\frac{dy}{dx}=x+y^2$, $y(0)=1$ by using Picard's method.

24. Write a FORTRAN program to compute $y(0.2)$ by Runge-Kutta method of order for the differential equation: $\frac{dy}{dx}=xy+y^2, y(0)=1$.

25. Write a FORTRAN program to find the roots of the equation $\cos x-xe^x=0$ up to 4-decimal places by using Newton-Rapson's method.

26. Write a FORTRAN program to minimize the objective function

Max $[15x+18y]$

S.t $5x+3y\geq2$

$3x+6y\geq3$

$x,y\geq0$

27. Write a FORTRAN program to solve the system following system of equations using Gauss-Seidal method. $27x+6y+z=85$; $6x+15y+2z=72$; $x+y+54z=110$

28. Consider the differential equation $\frac{dy}{dx}=x^2+1;\ y(0)=0.5$. Write a FORTRAN program to find the solution of the equation for $x=0\ to\ x=2$ by Euler and modified Euler methods.